

# Disproof of conjecture 00000001485: the cat-map suspension has zeta radius at most 1/2, not 1

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## 1 The conjecture and the clause refuted

**Statement (English, verbatim).** Definition: The zeta function of a suspension of an invertible map is the generating function  $\zeta(z) = \prod(1 - z^p)^{-1}$  of its periodic counts. Conjecture: For every hyperbolic suspension,  $\zeta$  has radius of convergence 1, and its singular set on the unit circle is exactly the set of roots of unity (a complete characterization of periodic structure); no hyperbolic suspension has radius of convergence  $> 1$ .

The conjecture is a conjunction. **We refute only its first clause**, “for every hyperbolic suspension,  $\zeta$  has radius of convergence 1”. That suffices to refute the conjunction. The witness is the suspension flow of Arnold’s cat map on the 2-torus. For it we prove that the radius of convergence of  $\zeta$  is at most 1/2. Its true value is  $(3 - \sqrt{5})/2$ , but only the bound is proved in Lean. The clause about the singular set is not formalized (see Remark 16). Neither is the clause “no hyperbolic suspension has radius  $> 1$ ”, which the example satisfies.

## 2 Reading and definitions (as formalized)

All objects below are Lean/Mathlib definitions in `Conjecture1485/Basic.lean`, namespace `C1485`. Here  $\mathbb{N} = \{0, 1, 2, \dots\}$ .

**Definition 1** (Torus and toral maps).  $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$  is the quotient of the additive group  $\mathbb{R}^2 = \text{Fin } 2 \rightarrow \mathbb{R}$  by the lattice `Lat`, the image of  $\mathbb{Z}^2$  (`Torus`). An integer matrix  $M$  induces the group endomorphism `tmap M`,  $x \bmod \mathbb{Z}^2 \mapsto Mx \bmod \mathbb{Z}^2$ .

**Definition 2** (Hyperbolic toral automorphism). `IsHyperbolic M` holds when  $\det M$  is a unit of  $\mathbb{Z}$  (so  $M \in GL(2, \mathbb{Z})$ ) and every complex eigenvalue  $\mu$  of  $M$  (`Module.End.HasEigenvalue of Matrix.toLin'`) has  $|\mu| \neq 1$ .

**Definition 3** (Cat map).  $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ , with inverse  $\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$ . The cat map `cat` is `tmap A`, packaged as an additive automorphism of  $\mathbb{T}^2$  with inverse `tmap Ainv`. So it is an invertible map.

**Definition 4** (Suspension and suspension flow). For a bijection  $e$  of a set  $\alpha$ , `Suspension e` is the quotient of  $\alpha \times \mathbb{R}$  by  $(x, t) \sim (e^k x, t - k)$  for  $k \in \mathbb{Z}$ . This is the equivalence relation generated by  $(x, t + 1) \sim (e x, t)$ , i.e. the mapping torus with unit roof. The flow is  $\varphi_s[x, t] = [x, t + s]$  (`flow`).

**Definition 5** (Periodic orbits, periodic counts,  $\zeta$ , radius). For  $f : \alpha \rightarrow \alpha$ , a *periodic orbit* (`PerOrbit f`) is a set  $\{f^k x : k \in \mathbb{N}\}$  with  $x$  a periodic point. Its period  $|\tau|$  (`PerOrbit.len`) is its number

of points, and it equals the least period of  $x$  (`PerOrbit.len_eq`). The periodic counts are  $N_n = \# \text{Fix}(f^n)$ . The zeta function is the infinite product over all periodic orbits,

$$\zeta_f(z) = \prod_{\tau} (1 - z^{|\tau|})^{-1} \in \mathbb{R}[[z]] \quad (\text{zeta f} := \text{tprod } (\text{fun tau} \Rightarrow \text{fac tau.len})),$$

with the Euler factor `fac`  $p = (1 - z^p)^{-1}$  (Mathlib's power-series inverse). The product is taken in Mathlib's coefficientwise topology on  $\mathbb{R}[[z]]$  (`PowerSeries.WithPiTopology`). Each orbit contributes its own factor, so the periodic counts enter with multiplicity. The radius of convergence `zetaRadius f` is Mathlib's `FormalMultilinearSeries.radius` of  $\sum_n \zeta_n z^n$  (`ofScalars` over  $\mathbb{C}$ ), i.e.  $\sup\{r \geq 0 : (\zeta_n |r^n)_n \text{ bounded}\}$ .

**Definition 6** (The refuted clause). **Clause1**: for every integer matrix  $M$  with `IsHyperbolic M`, `zetaRadius (tmap M) = 1`. This is the first clause, restricted to suspensions of hyperbolic automorphisms of  $\mathbb{T}^2$ . That is a subclass of all hyperbolic suspensions, so refuting it there refutes the “for every” claim.

### 3 Proof

**Lemma 7** (`A_hyperbolic`). *A is hyperbolic.*

*Proof.*  $\det A = 1$ . An eigenvalue  $\mu$  is a root of the characteristic polynomial  $z^2 - 3z + 1$  (`charpoly_fin_two`). Write  $\mu = a + bi$  with  $a^2 + b^2 = 1$ . The real and imaginary parts give  $a^2 - b^2 - 3a + 1 = 0$  and  $b(2a - 3) = 0$ , hence  $a(2a - 3) = 0$ . If  $a = 0$  then  $b = 0$ , contradicting  $a^2 + b^2 = 1$ . If  $a = 3/2$  then  $a^2 > 1$ , again a contradiction.  $\square$

**Lemma 8** (`card_ker_tmap`). *If  $\det M \neq 0$ , the kernel of  $\text{tmap } M$  is finite with exactly  $|\det M|$  elements.*

*Proof.* Let  $\psi : \mathbb{Z}^2 \rightarrow \mathbb{T}^2$ ,  $\psi(v) = M^{-1}v \bmod \mathbb{Z}^2$ . Then  $\ker \psi = M\mathbb{Z}^2$ , and the range of  $\psi$  is  $\ker(\text{tmap } M)$ : if  $Mx \in \mathbb{Z}^2$  then  $x = M^{-1}(Mx)$ . So  $\ker(\text{tmap } M) \cong \mathbb{Z}^2/M\mathbb{Z}^2$ . Mathlib's `Submodule.natAbs_det_equiv`, applied to the injective map  $v \mapsto Mv$ , gives  $\#(\mathbb{Z}^2/M\mathbb{Z}^2) = |\det M|$ .  $\square$

**Lemma 9** (`natAbs_det_ge`, `fix_cat`). *For  $n \geq 1$ ,  $\text{Fix}(\text{cat}^n)$  is finite and  $N_n = |\det(A^n - I)| \geq 2^n - 1$ .*

*Proof.*  $\text{cat}^n = \text{tmap}(A^n)$  (`cat_iterate`), so  $\text{Fix}(\text{cat}^n) = \ker \text{tmap}(A^n - I)$ . For  $2 \times 2$  matrices,  $\det(B - I) = \det B - \text{tr } B + 1$ , and  $\det A^n = 1$ , so  $\det(A^n - I) = 2 - t_n$  with  $t_n = \text{tr } A^n$ . From  $A^2 = 3A - I$  we get  $t_{n+2} = 3t_{n+1} - t_n$  (`trace_rec`), with  $t_0 = 2$  and  $t_1 = 3$ . By induction,  $2t_{n+1} \leq t_{n+2}$  and  $t_{n+1} \geq 2^{n+1} + 1$  (`trace_growth`). Hence  $|\det(A^n - I)| = t_n - 2 \geq 2^n - 1 > 0$ , and the previous lemma applies.  $\square$

**Lemma 10** (`flow_periodic_iff`). *For  $s > 0$ :  $\varphi_s[x, t] = [x, t]$  iff  $s = n$  for some integer  $n > 0$  with  $e^n x = x$ .*

*Proof.*  $[x, t + s] = [x, t]$  iff there is  $k \in \mathbb{Z}$  with  $x = e^k x$  and  $t = t + s - k$ . That is,  $s = k$ , and  $k > 0$  because  $s > 0$ .  $\square$

So the closed orbits of the suspension flow are exactly the flow lines through the periodic points of  $e$ , and the least period of the flow line through  $[x, t]$  is the least period of  $x$  under  $e$ . The periodic orbits and periods of the suspension are therefore those of the cat map, which is how  $\zeta$  is built.

**Lemma 11** (`finite_orbits`, `hasProd_zeta`). *If every  $\text{Fix}(f^m)$  with  $m \geq 1$  is finite, then for each  $k$  there are finitely many orbits with  $|\tau| \leq k$  (call this set  $F_k$ ). The product defining  $\zeta_f$  converges (`HasProd`), and  $\zeta_k$  equals the  $z^k$ -coefficient of the finite product  $\prod_{\tau \in F_k} (1 - z^{|\tau|})^{-1}$ .*

*Proof.* For  $p > k$ ,  $(1 - z^p)^{-1} - 1 = z^p(1 - z^p)^{-1}$  is divisible by  $z^{k+1}$ , and a product of series that are  $\equiv 1 \pmod{z^{k+1}}$  is again  $\equiv 1$ . Hence for every finite set  $S \supseteq F_k$  of orbits, the  $z^k$ -coefficient of  $\prod_{\tau \in S}$  equals that of  $\prod_{\tau \in F_k}$ . So the partial products converge coefficientwise, and `tprod` is their limit.  $\square$

**Lemma 12** (`fac_eq`, `prod_ge`, `coeff_zeta_ge`). *For  $k \geq 1$ ,  $\zeta_k \geq \#\{\tau \in F_k : |\tau| \text{ divides } k\}$ .*

*Proof.*  $(1 - z^p)^{-1} = \sum_{j \geq 0} z^{pj}$  for  $p \geq 1$ , so each factor  $G_\tau$  has  $G_\tau - 1$  with nonnegative coefficients. By induction over a finite set,  $\prod G_\tau - 1 - \sum (G_\tau - 1)$  and  $\prod G_\tau - 1$  have nonnegative coefficients. Taking the  $z^k$ -coefficient gives  $\zeta_k \geq \sum_{\tau \in F_k} [\tau] \mid k$ .  $\square$

**Lemma 13** (`fix_le_orbits`). *For  $k \geq 1$ ,  $\#\text{Fix}(f^k) \leq k \cdot \#\{\tau \in F_k : |\tau| \mid k\}$ .*

*Proof.* Each  $x \in \text{Fix}(f^k)$  is periodic, and its orbit  $\tau$  has  $|\tau| \mid k$ , so  $\tau \in F_k$ . The orbit map  $x \mapsto \tau$  has fibres contained in  $\{f^j x_0 : j < k\}$ , which have at most  $k$  points.  $\square$

**Lemma 14** (`radius_le_half`). *If  $nc_n \geq 2^n - 1$  for all  $n \geq 1$ , the series  $\sum c_n z^n$  has radius  $\leq 1/2$ .*

*Proof.* Suppose the radius exceeds  $1/2$ . Pick  $r$  with  $1/2 < r < \min(\text{radius}, 1)$ . Then  $|c_n| r^n \leq C$  for all  $n$  (`norm_mul_pow_le_of_lt_radius`), so  $(2r)^n = 2^n r^n \leq (nc_n + 1)r^n \leq nC + 1$ . This contradicts  $n/(2r)^n \rightarrow 0$ , since  $2r > 1$ .  $\square$

**Theorem 15** (`conjecture_00000001485_false`). *A is hyperbolic. For all  $n \geq 1$ ,  $\#\text{Fix}(\text{cat}^n) = |\det(A^n - I)|$ . The closed orbits of the suspension flow of `cat` are characterized as in `flow_periodic_iff`. The product  $\prod_\tau (1 - z^{|\tau|})^{-1}$  converges to  $\zeta_{\text{cat}}$ . And `zetaRadius cat`  $\leq 1/2$ , hence  $\neg \text{Clause1}$ .*

*Proof.* Combining the lemmas, for  $n \geq 1$ :  $2^n - 1 \leq N_n \leq n \cdot \#\{\tau \in F_n : |\tau| \mid n\} \leq n \zeta_n$ . So `radius_le_half` applies. If `Clause1` held, then applying it to  $M = A$  would give radius  $1 \leq 1/2$ , which is false.  $\square$

**Remark 16** (Scope and readings). (i) Only clause 1 is refuted. The singular-set clause and the clause “no radius  $> 1$ ” are not addressed in Lean. As a prose remark only, not formalized: clause 2 cannot hold for any power series of radius 1. The set of points on the circle of convergence where the series has no analytic continuation is closed, while the set of roots of unity is not closed. (ii) “Hyperbolic suspension” is instantiated by the subclass of suspensions of hyperbolic automorphisms of  $\mathbb{T}^2$ . General Anosov flows are not formalized. (iii)  $\prod (1 - z^p)^{-1}$  is read as a product over periodic orbits, so the periodic counts appear with multiplicity. This is equivalent to  $\exp(\sum_n N_n z^n / n)$ ; that equivalence is not used or formalized. The alternative reading, a product over the set of periods without multiplicity, is not refuted. It does not encode any periodic counts. (iv) Conventions: `Mathlib's Function.periodicPts` requires a positive period, so every  $|\tau| \geq 1$  (`PerOrbit.len_pos`) and every Euler factor is a genuine inverse.

## 4 Relation to the earlier submission

The earlier submission is PR #262 by orionsheep, closed without merging. The reviewer wrote (Unicode symbols transliterated): “conjecture\_refuted is a conjunction of decidable trivia (four periodic counts,  $\mathbb{Z}[\sqrt{5}]$  identities, sign flags) that never mentions  $\zeta$ , the suspension, or a radius of convergence. The cat-map counterexample — valid in prose — is entirely unformalized, and only finitely many periodic counts are certified where the radius of convergence needs all of them.”

This submission uses the same counterexample, and formalizes everything that the earlier Lean lacked:

- the torus  $\mathbb{R}^2/\mathbb{Z}^2$ , with the cat map as an actual automorphism of it, and a proof of hyperbolicity;
- the periodic counts  $\# \text{Fix}(\text{cat}^n) = |\det(A^n - I)|$  for *every*  $n \geq 1$ ;
- the suspension flow and its closed orbits;
- $\zeta$  as the convergent infinite product over all periodic orbits;
- Mathlib’s radius of convergence of  $\zeta$ , bounded using all coefficients.

## 5 Lean formalization and axiom audit

Package: `lean/Conjecture1485/Basic.lean` (445 lines, `import Mathlib` only; Lean v4.33.1, Mathlib v4.33.1), identical to the file built with `lake build`. The main theorem is `C1485.conjecture_00000001485_false`. The lemmas are named in the sections above. `#print axioms C1485.conjecture_00000001485_false` reports `[propext, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide`. `decide` is used only for  $2 \times 2$  integer matrix identities ( $AA^{-1} = I$ ,  $A^2 = 3A - I$ ,  $\det A = 1$ ,  $\text{tr } A^0 = 2$ ,  $\text{tr } A = 3$ ).

## Sources

- Earlier submission and review: <https://github.com/The-Last-Math-Competition/The-Last-Math-Competition/pull/262>.
- Mathlib v4.33.1 (definitions and the theorems named above, e.g. `Mathlib/LinearAlgebra/FreeModule/Finite/CardQuotient.lean`).